

association-analysis

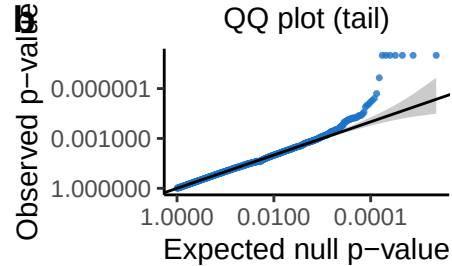
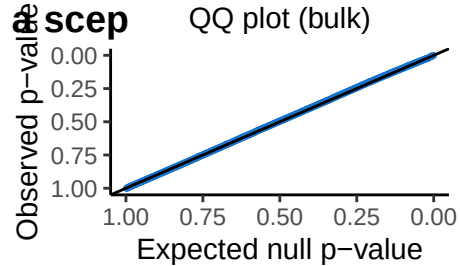
2026-04-02

REPLOGLE

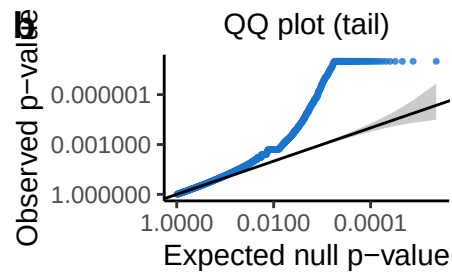
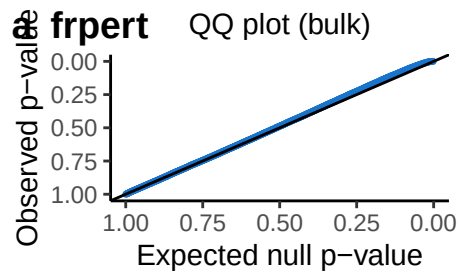
STATISTICAL

```
## Dataset: replogle-rd7_pos_ctrl_ntargets=1500_ncells=100k_gene_thresh=7
##   #genes = 1500 ; #cells = 100000
##   #guides = 1716 ; #targets = 1501 ; avg # guides / target: 1.14
##   #cells per guide: median = 47 ; min = 1 ; max = 1124
##   #cells per target: avg = 48 ; min = 16 ; max = 5962
##   #guides per cell: moi (avg) = 1 ; min = 1 ; max = 1

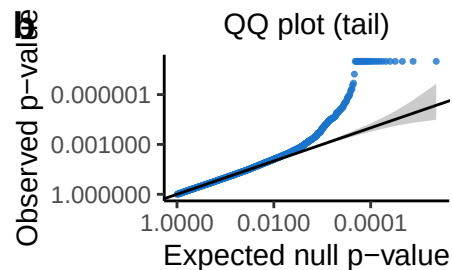
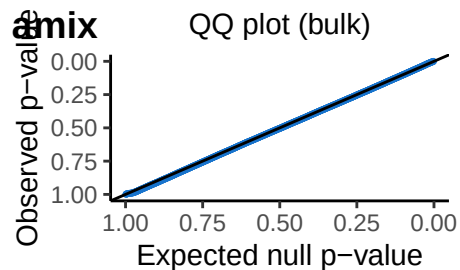
## Dataset: replogle-rd7_neg_ctrl_ngenes=1000_gene_thresh=7
##   #genes = 1000 ; #cells = 13548
##   #guides = 113 ; #targets = 113 ; avg # guides / target: 1
##   #cells per guide: median = 111 ; min = 25 ; max = 619
##   #cells per target: avg = 111 ; min = 25 ; max = 619
##   #guides per cell: moi (avg) = 1 ; min = 1 ; max = 1
##   #discovery pairs: 113000
```



Num. signif
(at alpha 0.1):
15 of 113000

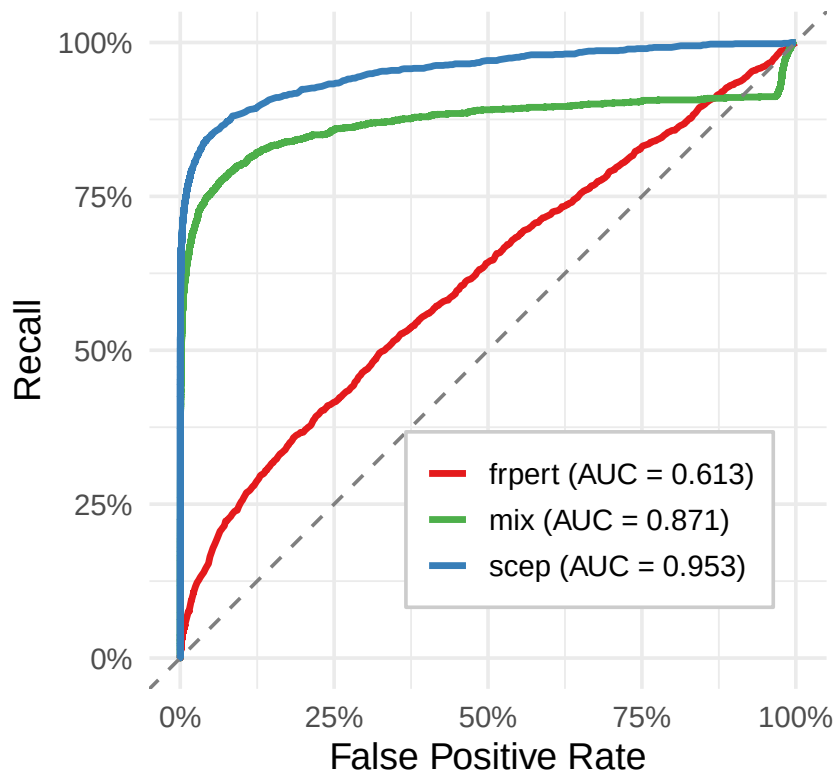


Num. signif
(at alpha 0.1):
596 of 113000

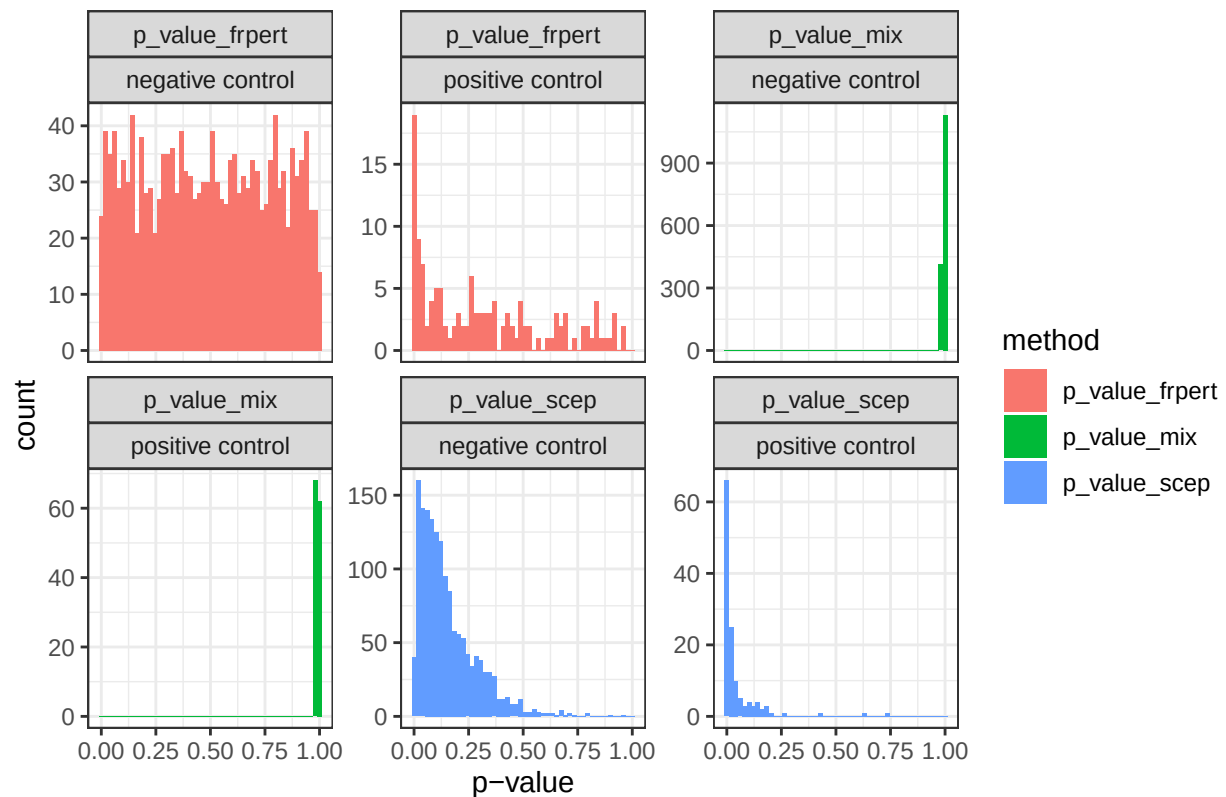


Num. signif
(at alpha 0.1):
107 of 113000

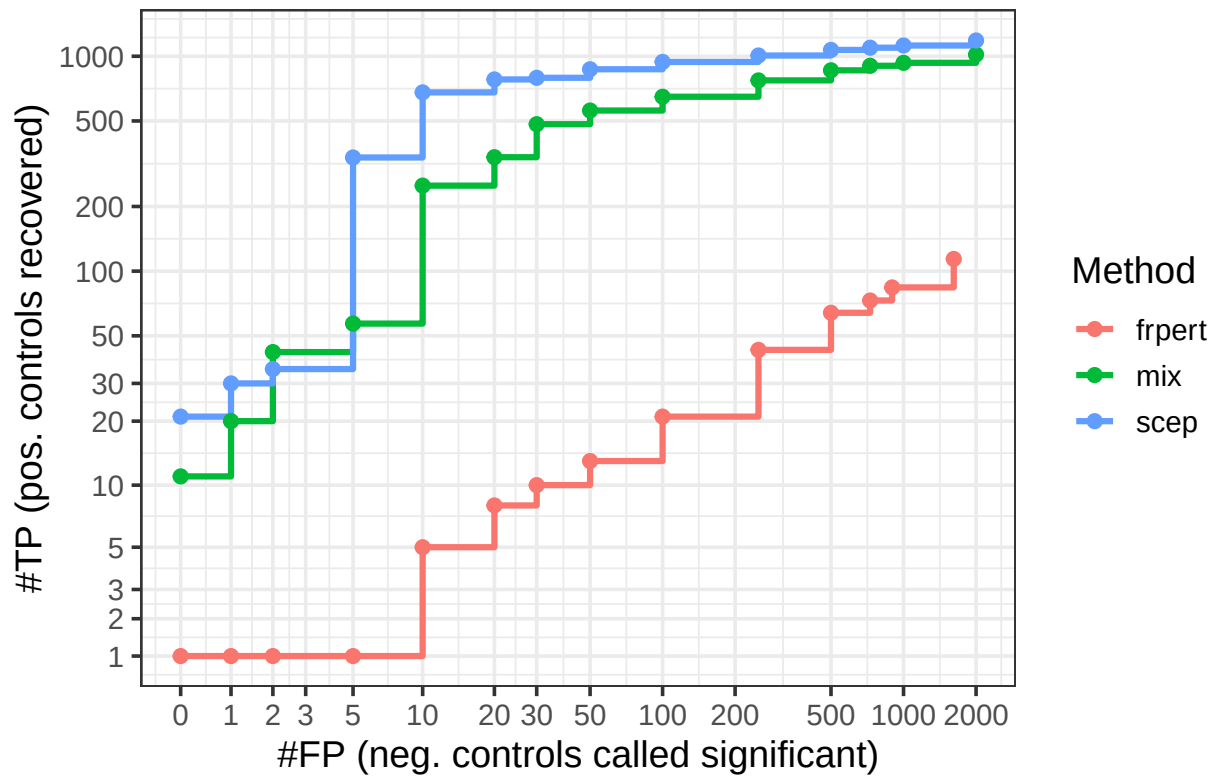
Replogle: Separation of pos. and neg.



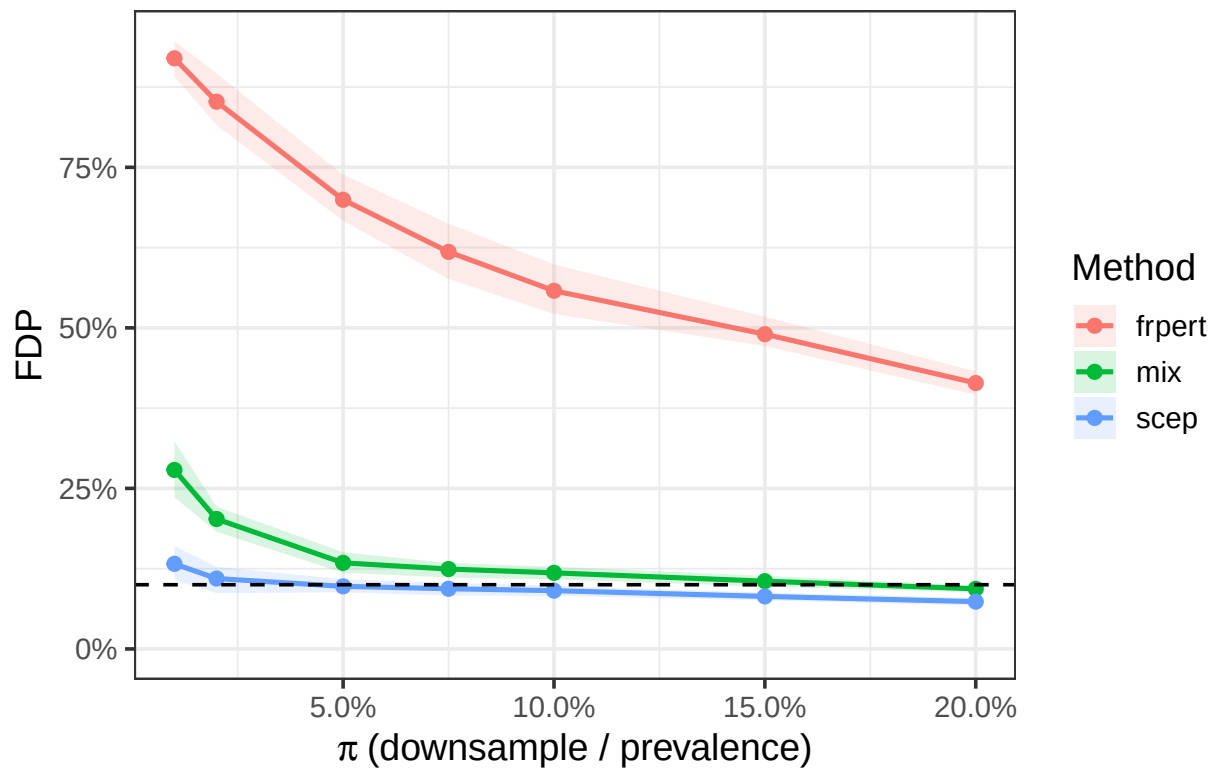
Replogle: All pairs with $n_{\text{nonzero_trt}} = 0$ (no QC)

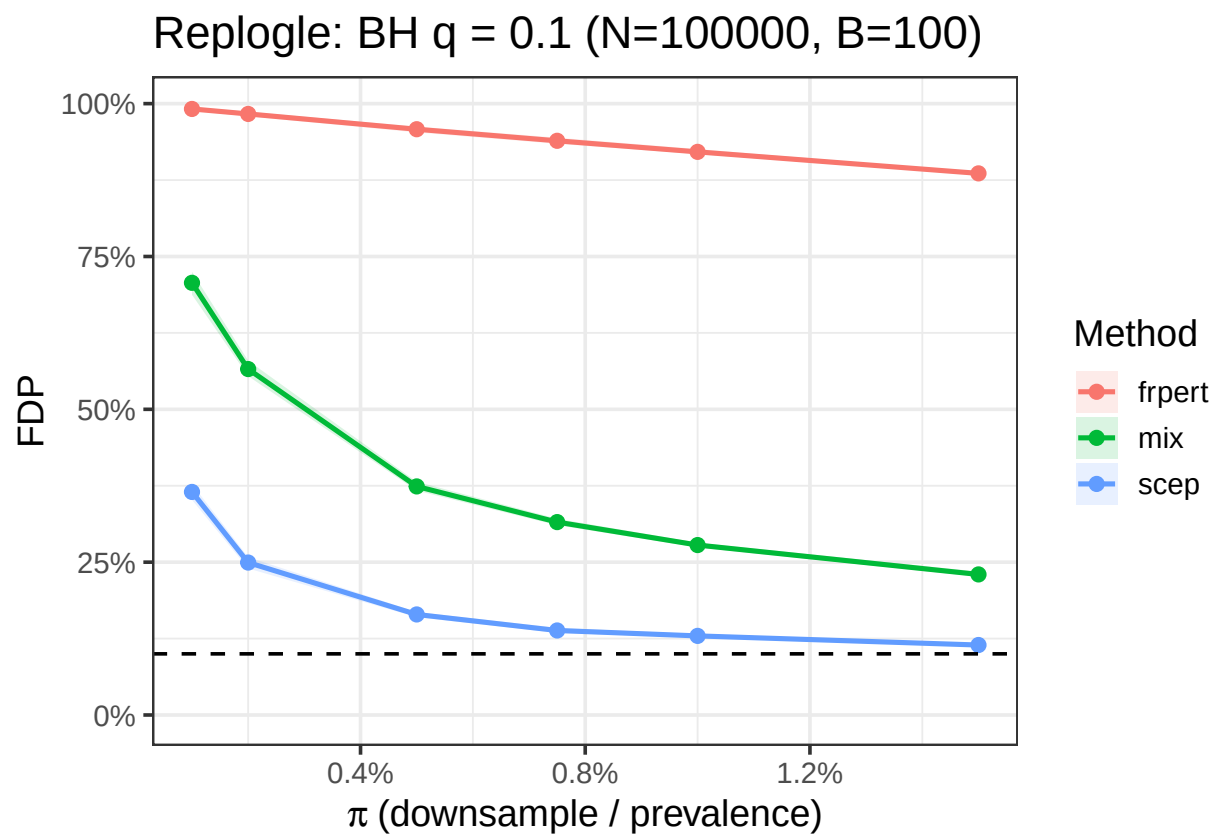


Replogle: #TP recovered vs #FP allowed



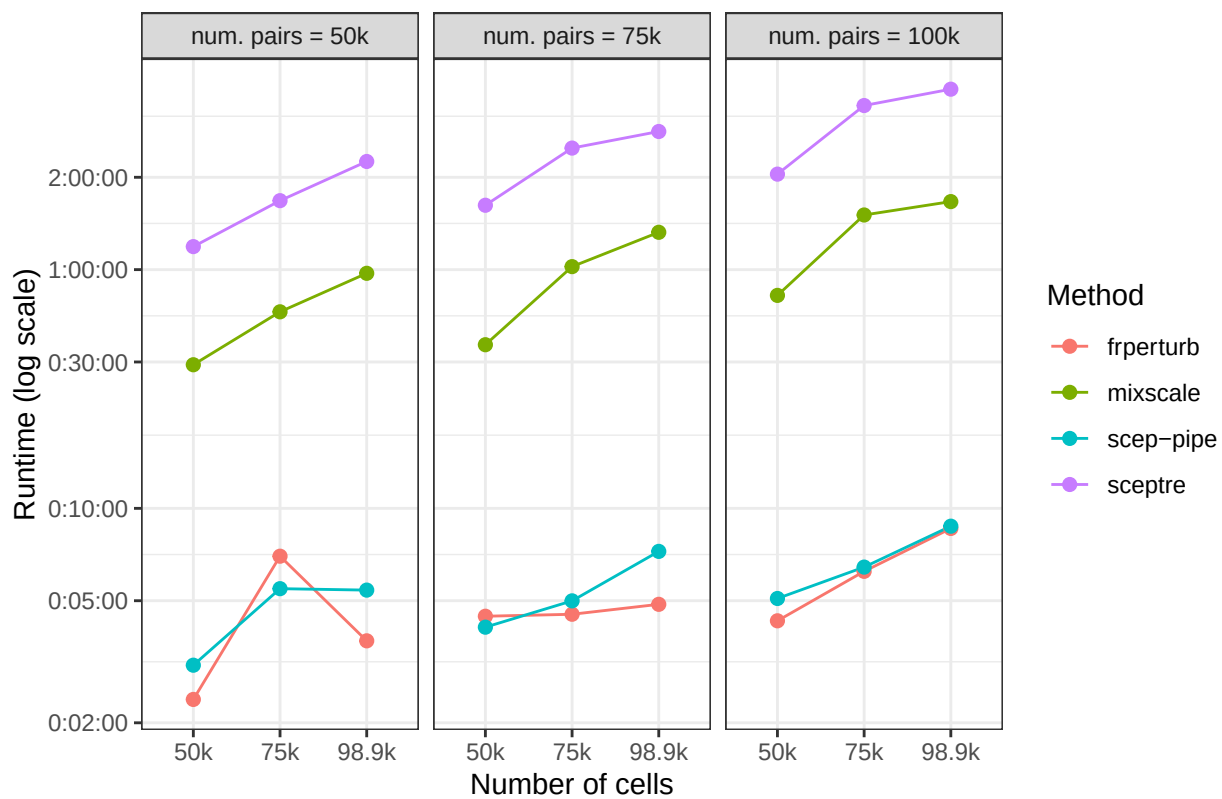
Replogle: BH $q = 0.1$ (N=7500, B=100)

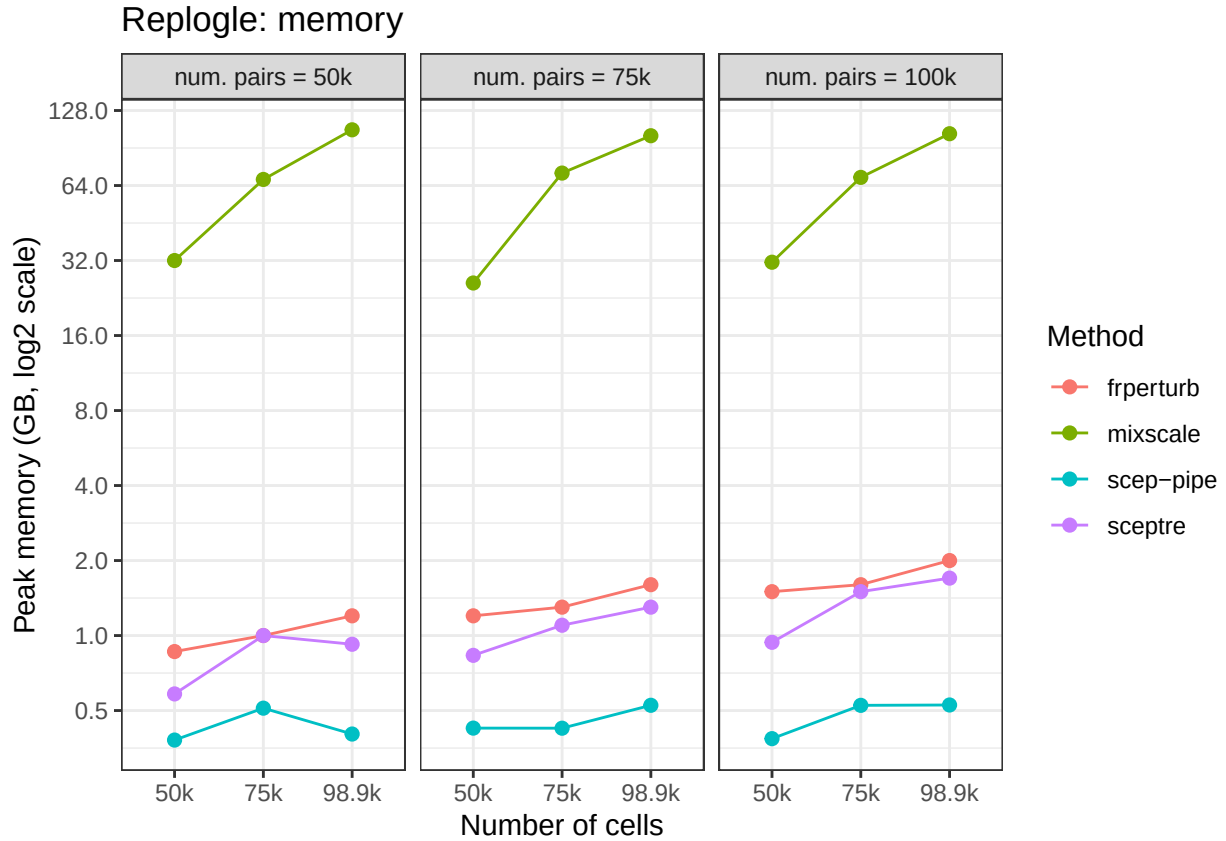




2. COMPUTATIONAL

Replay: run time



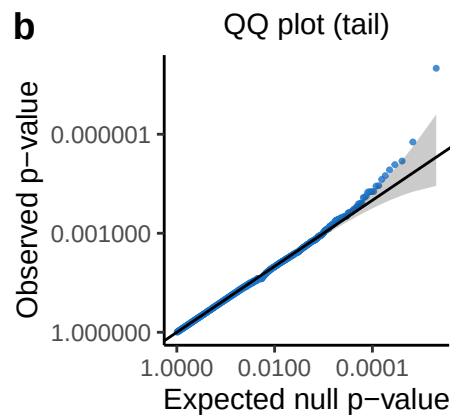
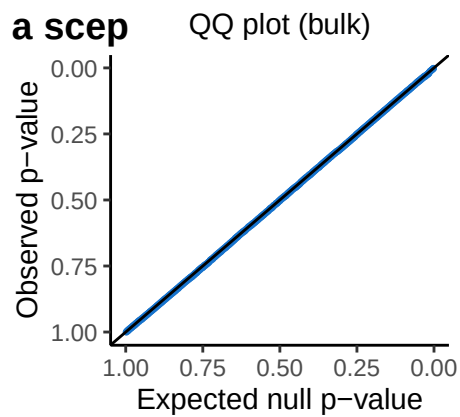


TODO: sceptre-pipeline extras

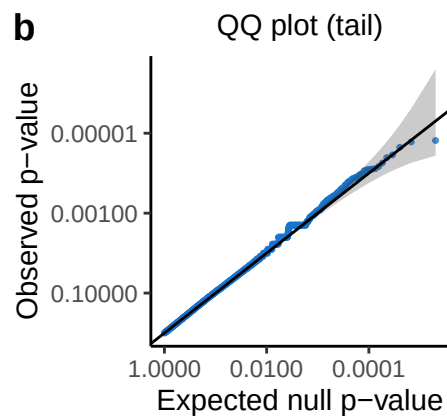
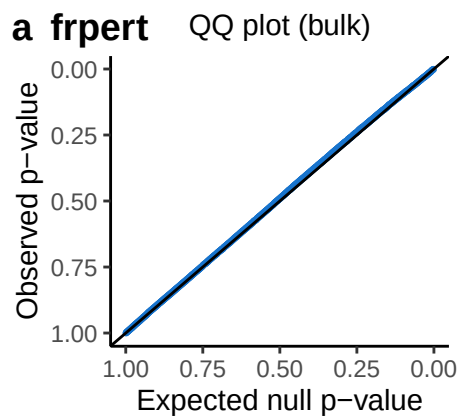
2. GASPERINI

```
## Dataset: gasperini_pos_ctrl_ntargets=377_ncells=100k_gene_thresh=7
## #genes = 377 ; #cells = 100000
## #guides = 855 ; #targets = 378 ; avg # guides / target: 2.26
## #cells per guide: median = 362 ; min = 40 ; max = 749
## #cells per target: avg = 747 ; min = 196 ; max = 31085
## #guides per cell: moi (avg) = 3.13826 ; min = 1 ; max = 22

## Dataset: gasperini_neg_ctrl_ngenes=1000_gene_thresh=7
## #genes = 1000 ; #cells = 39323
## #guides = 101 ; #targets = 101 ; avg # guides / target: 1
## #cells per guide: median = 444 ; min = 182 ; max = 937
## #cells per target: avg = 444 ; min = 182 ; max = 937
## #guides per cell: moi (avg) = 1.165603 ; min = 1 ; max = 6
## #discovery pairs: 101000
```

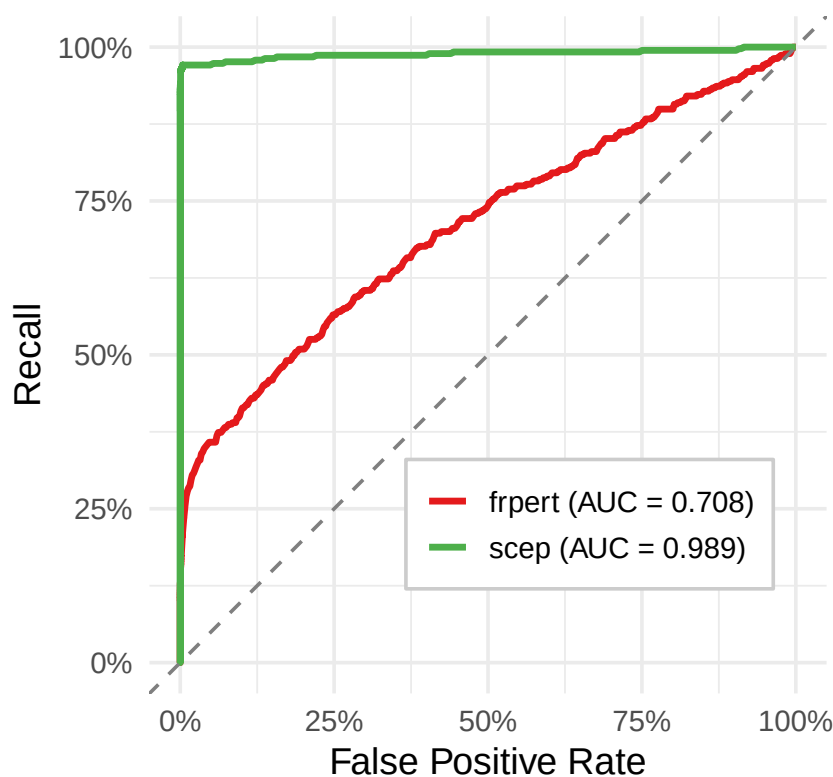


Num. signif
(at alpha 0.1):
2 of 101000

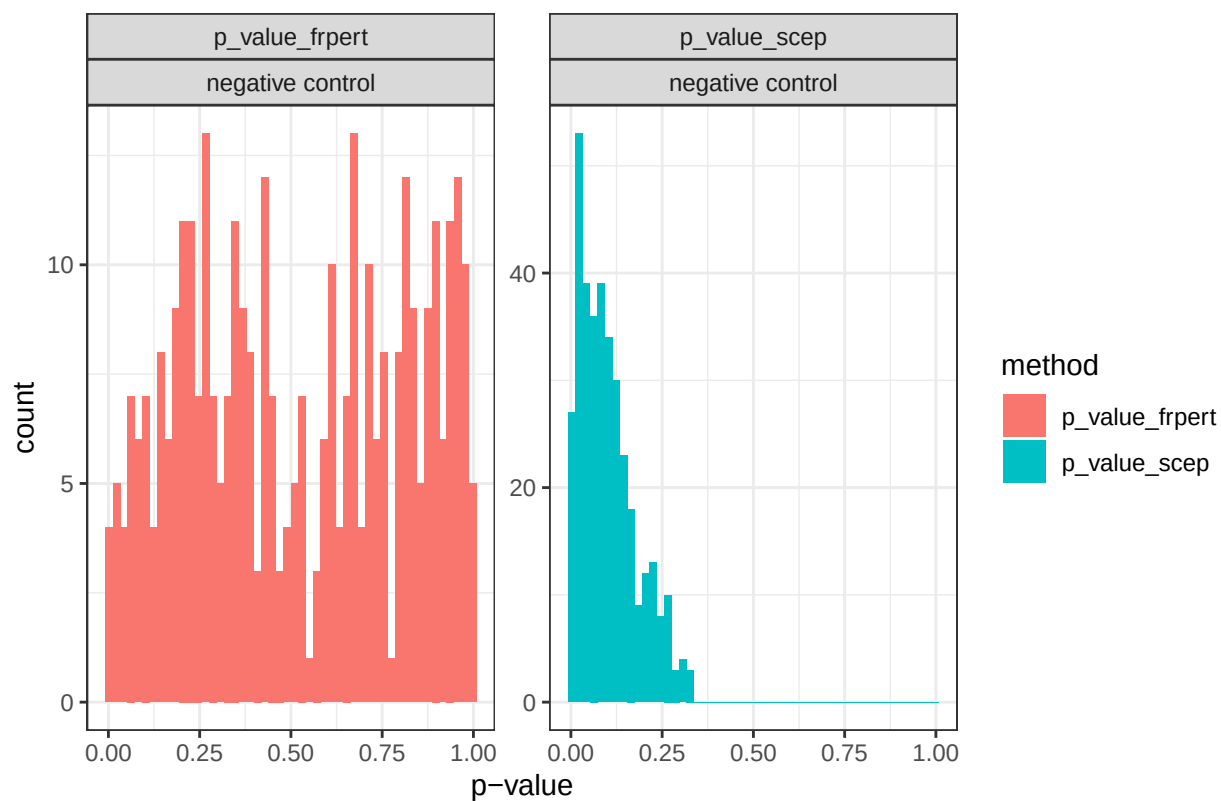


Num. signif
(at alpha 0.1):
0 of 101000

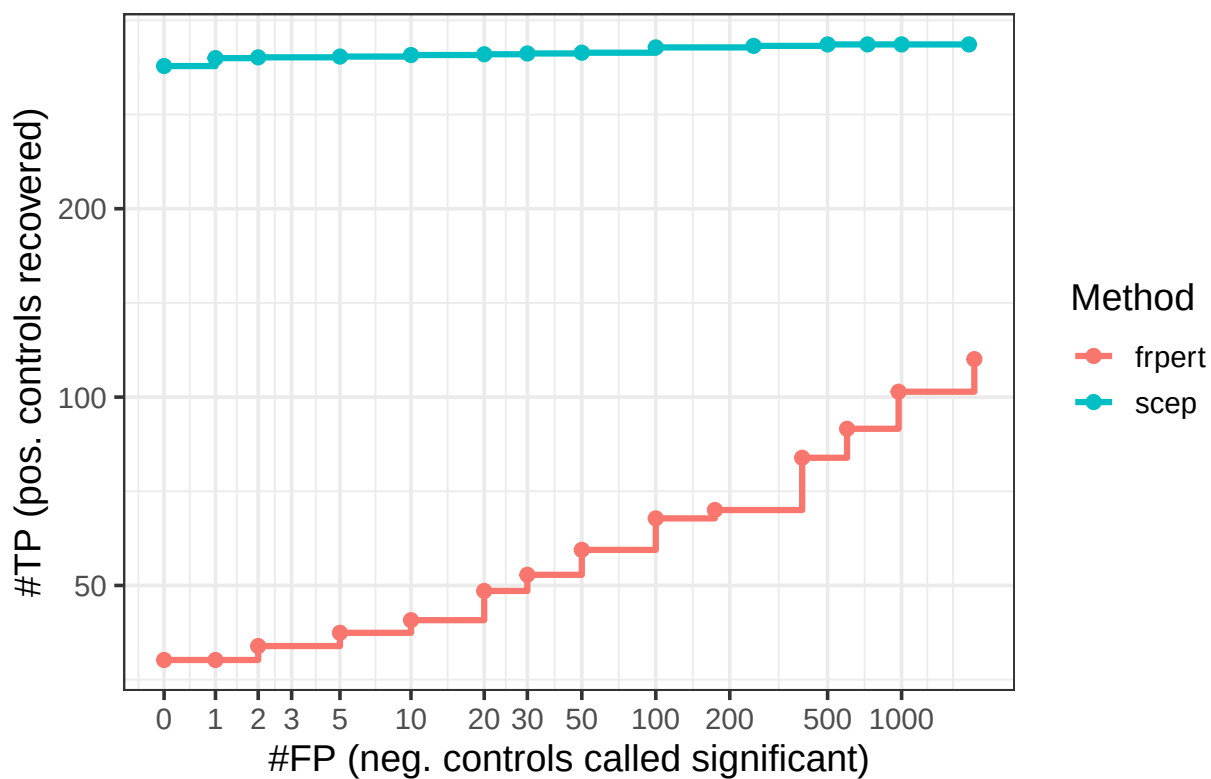
Gasparini: Separation of pos. and neg.



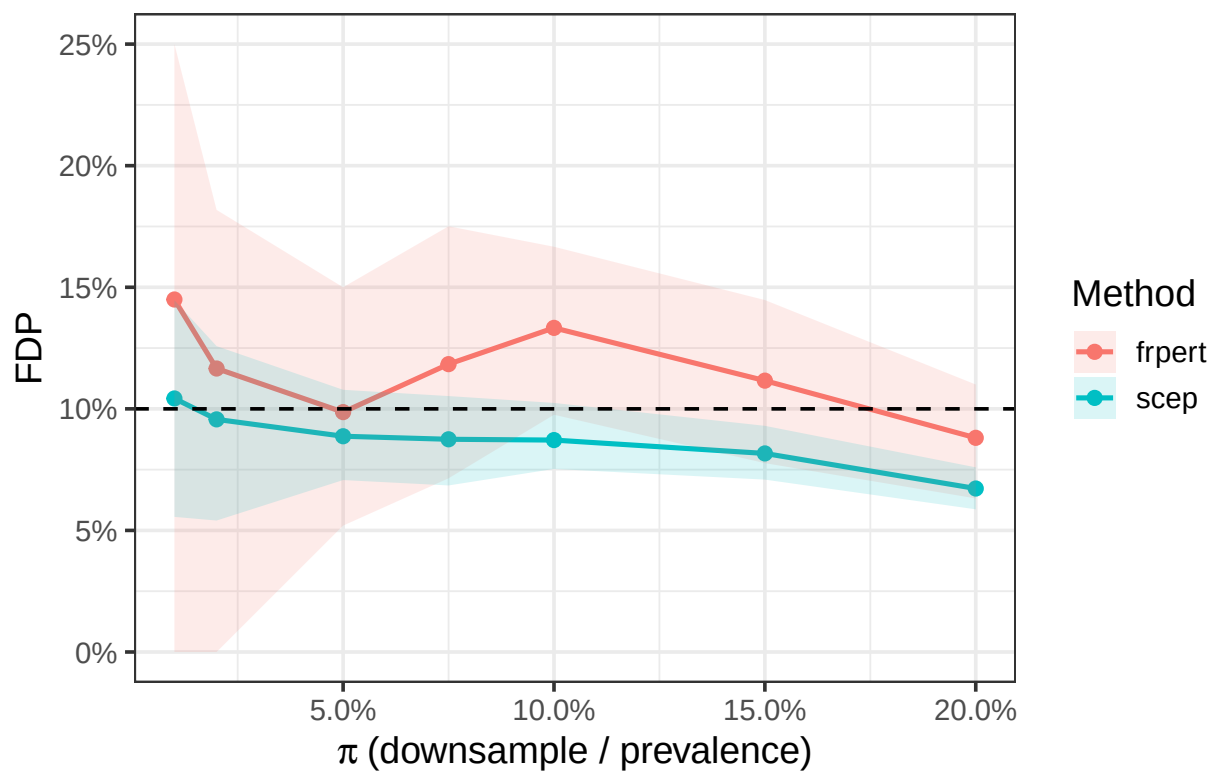
Gasperini: All pairs with $n_{\text{nonzero_trt}} = 0$ (no QC)



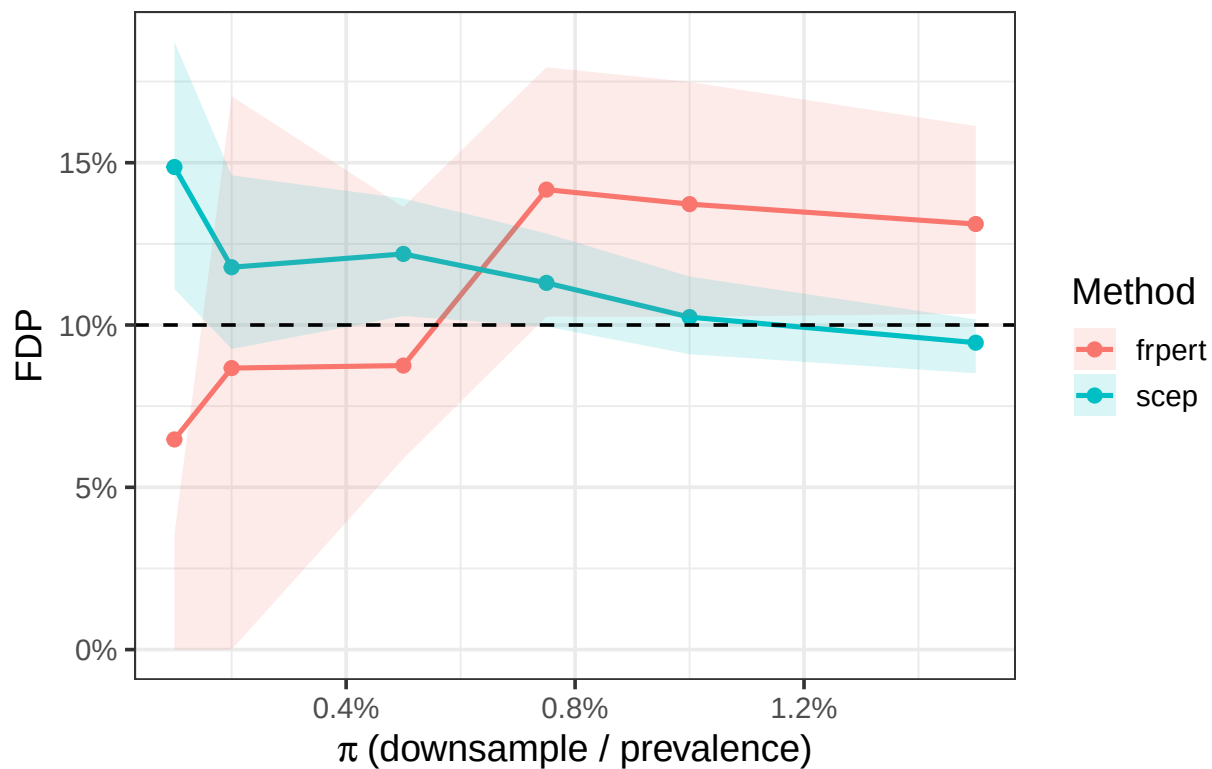
Gasperini: #TP recovered vs #FP allowed



Gasperini: BH $q = 0.1$ (N=1880, B=100)

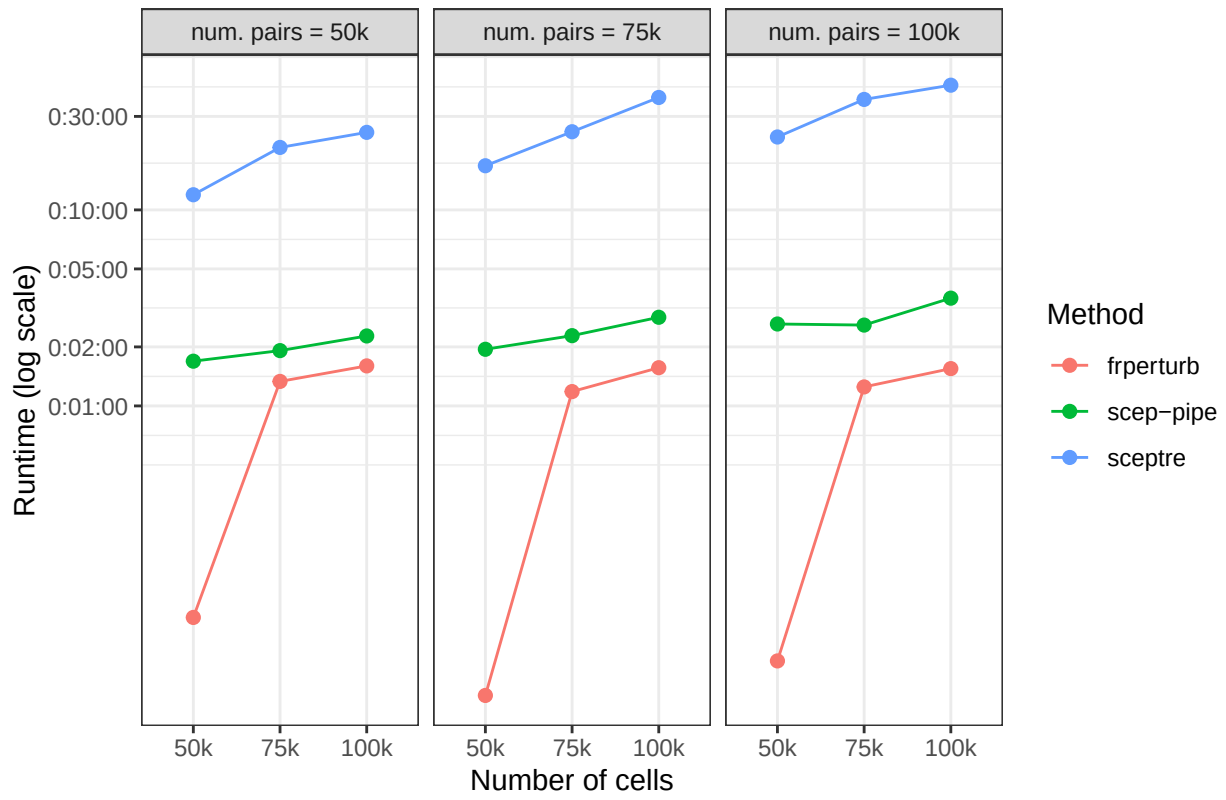


Gasperini: BH $q = 0.1$ (N=25100, B=100)



COMPUTATIONAL

Gasperini: run time



Gasperini: memory

